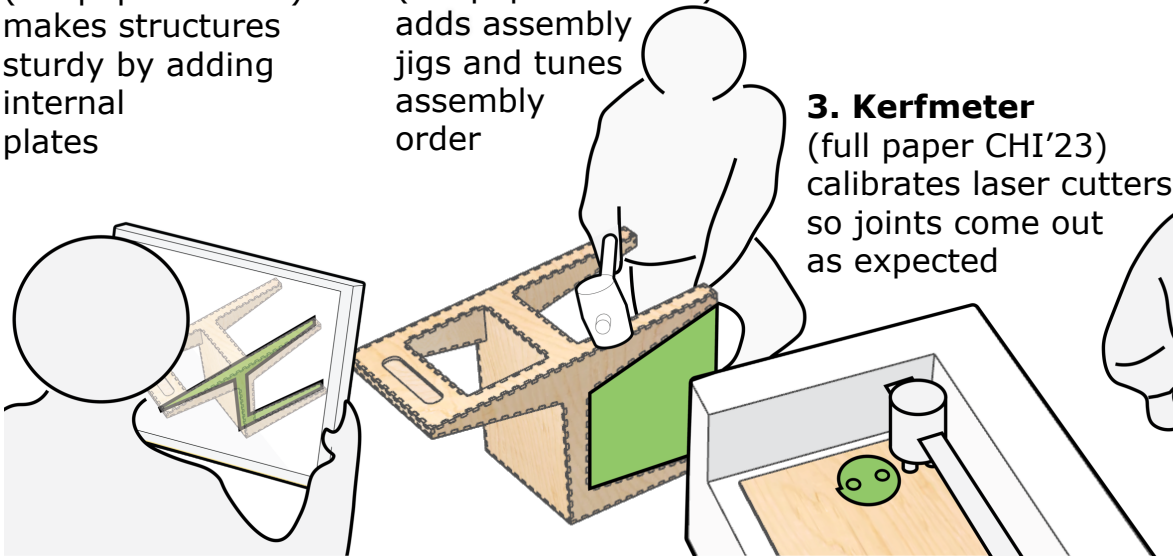
status quo:
requires expert software and expert knowledge



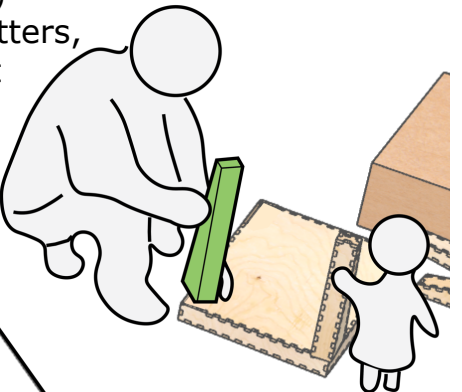
1. FastForce
(full paper CHI'21)
makes structures sturdy by adding internal plates

2. Mallet-based assembly
(full paper UIST'25)
adds assembly jigs and tunes assembly order

3. Kerfmeter
(full paper CHI'23)
calibrates laser cutters, so joints come out as expected



4. PÅCKER
(full paper UIST'26)
makes objectstrivial to assemble —IKEA style



100
experts can do it

100,000
non-experts can do it

100,000,000
consumers can do it